

Feb 20, 2026

Timur / Jesus - Transcript

00:00:00

Timur Khamidulin: Hello. Hello Jesus. All good. All good. How are you? You're without camera today. H. You're without it. Yeah. Okay. Okay.

Jesus Gonzalez Godoy: Um so also I prefer turn on our cameras but also in the future meetings if you are more comfortable without camera.

Timur Khamidulin: No, I'm fine with I'm fine with I'm fine with the camera.

Jesus Gonzalez Godoy: Okay. Okay. Okay. Good.

Timur Khamidulin: It's just our our first meeting without camera.

Jesus Gonzalez Godoy: Um

Timur Khamidulin: That's

Jesus Gonzalez Godoy: yeah it's difficult for me today.

Timur Khamidulin: why

Jesus Gonzalez Godoy: Um, you have finished the the project.

Timur Khamidulin: yes, I have finished the project. The only thing that uh I am still doing the evaluation uh like not still doing I mean like I need to do some uh additional work on evaluation but the pipeline is ready and uh so it works uh if you if I start the pipeline the pipeline works and I can uh put queries there and it um shows me the results.

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Timur Khamidulin: So, so the pipeline works.

Jesus Gonzalez Godoy: Okay. So, can we check

Timur Khamidulin: Yes. Um, one second. So, So yes, let me show my screen. Can you see? Okay.

Jesus Gonzalez Godoy: Yes.

Timur Khamidulin: So uh just uh the Oh yes. So as you said like all the uh so the pipeline I just like like uh as as as as we discussed just copy pasted uh the uh the uh what's it called like the the weeks from notebook to pipeline. So we can uh load it and um you can't see.

Jesus Gonzalez Godoy: Are you are you sharing that because

Timur Khamidulin: uh let's do maybe uh more again. Wait. And uh what is

Jesus Gonzalez Godoy: Okay.

Timur Khamidulin: uh uh here?

Jesus Gonzalez Godoy: So this is more like um

Timur Khamidulin: Yeah. And here and here is here's more more more

Jesus Gonzalez Godoy: Yeah. So first this is more like a like an app application, right?

More than a framework because you run the framework.py by and it runs the an interactive uh program or an interactive application, right? So, can we just go to the code? So, did you include a main? I think uh I think at the end should be it should have a a

Timur Khamidulin: So

Jesus Gonzalez Godoy: main. Yeah, main. So in that sense when you run you need to directory has on default.

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Jesus Gonzalez Godoy: Then first you run this uh this script and you got some errors right? What are those errors? Uh yeah the compressing data incorrect found object. Do you know what what are those errors?

Timur Khamidulin: Um, no because before there there was there was no such an error but I

Jesus Gonzalez Godoy: No.

Timur Khamidulin: couldn't wait again.

Jesus Gonzalez Godoy: So you create a monitor, you create the vector database. Could you scroll down a little bit? And then you create the rack. you send to the input to the rack instance is the LLM and vector database interactive mode. Okay, so it is it is okay. I just wanted to have this more in the notebook and you use this as um framework in the same way that you use for example TensorFlow or pandas but it's okay you created a interactive uh app from command line

Timur Khamidulin: I I Yeah,

Jesus Gonzalez Godoy: Um

Timur Khamidulin: I thought that uh Ah, you want you wanted me so we could do it in a notebook.

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Timur Khamidulin: You wanted

Jesus Gonzalez Godoy: yes but this is this way is okay but the other thing is just remove the main and do it in

Timur Khamidulin: this.

Jesus Gonzalez Godoy: the notebook but it's okay now that is the first thing the other thing is

Timur Khamidulin: You okay?

Jesus Gonzalez Godoy: um the errors let me see because she didn't I think these are something related with the vector database maybe. Can you comment everything

Timur Khamidulin: Okay,

Jesus Gonzalez Godoy: uh after initialize the vector database?

Timur Khamidulin: I like that.

Jesus Gonzalez Godoy: You can uh also select all the block and go to

Timur Khamidulin: like this. Wait, okay.

Jesus Gonzalez Godoy: Yeah.

Timur Khamidulin: No, not like this. And and just delete it.

Jesus Gonzalez Godoy: Now go to uh file.

Timur Khamidulin: Or or what?

Jesus Gonzalez Godoy: Let me see like this in selection something like to go to Google line command and edit in edit in the menu edit uh to line toggle block command. Yes, this is all the block and you can just please uh undo that that action.

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Jesus Gonzalez Godoy: Ctrl C. Yeah. And now again but to go line command. Now select to line command. Yes.

Timur Khamidulin: M okay.

Jesus Gonzalez Godoy: So these are the two different ways.

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: Um so just now please in comment

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: monitor.

Timur Khamidulin: I'm coming.

Jesus Gonzalez Godoy: Yes. Now run the script. Okay, there now command monitor. Okay. So in monitor is the error. So, but you you see the difference right between run

everything here in the command line

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: or just comment this main and do not put any entrance point of the script and just uh run this in the notebook. Uh can you see the difference or

Timur Khamidulin: So you mean that it's not uh so here is less uh understandable where the problem

Jesus Gonzalez Godoy: it's different purpose.

Timur Khamidulin: is.

Jesus Gonzalez Godoy: So one is if you want to what if you want to have an application that runs

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Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: on CLI uh this is the way you

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: put this main and then you run from terminal because you

Timur Khamidulin: Uh-huh.

Jesus Gonzalez Godoy: define this interactive loop this while true input and so on this interactive loop so you just um can interact with the

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: rack. So this is um like an application terminal application that you created here. The other idea is to have just the framework and just to have define the classes and so on and be able to create the instances and uh work with the rack but on the on another script or on the notebook or so on using this framework. That's the

Timur Khamidulin: Okay. So I So okay I think by Monday I can do it.

Jesus Gonzalez Godoy: difference.

Timur Khamidulin: So like by I can do the notebook and uh like

Jesus Gonzalez Godoy: Yeah.

Timur Khamidulin: uh

Jesus Gonzalez Godoy: So let's but it's okay. Uh the important thing is that you see clearly the difference

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Timur Khamidulin: okay.

Jesus Gonzalez Godoy: that one is like to use as a tool as a framework

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: independently and the other one that you created is a completely uh different purpose that is an application from terminal.

Timur Khamidulin: Mhm. So I need to do the notebook uh for

Jesus Gonzalez Godoy: Okay.

Timur Khamidulin: the

Jesus Gonzalez Godoy: Uh let's let's see uh what is the error on the

Timur Khamidulin: uh

Jesus Gonzalez Godoy: monitor? Can you open can you go to monitor? What is what is monitor?

Timur Khamidulin: Oh my god.

Jesus Gonzalez Godoy: It's It's up

Timur Khamidulin: Yes. One second.

Jesus Gonzalez Godoy: money tour. Yeah.

Timur Khamidulin: Okay. Yes.

Jesus Gonzalez Godoy: Did you create this code or you ask to Germany also create

Timur Khamidulin: Uh no for for the monitor I asked the uh the

Jesus Gonzalez Godoy: structure? Okay.

Timur Khamidulin: Germany to do it.

Jesus Gonzalez Godoy: Try also this is important for every class you created with AI um ask to include a description in doc strings.

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Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: Have you seen this description before?

Timur Khamidulin: Yes. Yes. Of course. Dog strings. Yes. Yes. Dog strings and markdowns. So markdowns is quite different. It's in a notebook. But do strings is like it goes in double uh it's it's like comment like we did here like here you remember like we put edit.

Jesus Gonzalez Godoy: Yes,

Timur Khamidulin: So yeah I understand.

Jesus Gonzalez Godoy: maybe maybe you can copy copy the monitor press paste it in

any or copy or just select what what AI are

Timur Khamidulin: Yes.

Jesus Gonzalez Godoy: you using to select and then make left right

Timur Khamidulin: Uh,

Jesus Gonzalez Godoy: click and where you have your AI assistant.

Timur Khamidulin: I can put uh in in cloth in uh

Jesus Gonzalez Godoy: Oh, you are using cloud.

Timur Khamidulin: I can use Yeah, I have I have an access to it. I can use a cloth and I can use Codex or I can just go to Gemini.

Jesus Gonzalez Godoy: Okay. So, okay.

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Jesus Gonzalez Godoy: So, let's see how how you how are you using the the AI? I mean because for Visual Studio you can make this interactive and you have a window where you can ask to modify the uh the selection.

Timur Khamidulin: I I regularly go I

Jesus Gonzalez Godoy: You have copilot for

Timur Khamidulin: I No,

Jesus Gonzalez Godoy: example.

Timur Khamidulin: I don't have copilot cuz I uh don't have access to it. It's uh it's only with subscri sub

Jesus Gonzalez Godoy: Okay.

Timur Khamidulin: subscription.

Jesus Gonzalez Godoy: And you have Germany

Timur Khamidulin: Uh yeah.

Jesus Gonzalez Godoy: Uh,

Timur Khamidulin: It's Jimmy.

Jesus Gonzalez Godoy: can you open cloud?

Timur Khamidulin: Uh, yes, of course.

Jesus Gonzalez Godoy: I haven't used cloud in.

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: Okay.

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: Uh, try to No,

Timur Khamidulin: I can like

Jesus Gonzalez Godoy: no, not no. Yeah, don't paste it.

Timur Khamidulin: this.

Jesus Gonzalez Godoy: Try to ask add dog strings for the selected code to see if it can do that. Yeah.

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Jesus Gonzalez Godoy: Okay. Yes. So the the difference or the idea for these doc strings is that when you publish your code you can easily generate documentation that helps um I don't know if you have seen in your notebooks that when you hover one class or something it says a legend that some some kind of help about the class. Have you seen that? Uh or for example,

Timur Khamidulin: No.

Jesus Gonzalez Godoy: can we go can you save that and can we go to the main where where did you create the monitor? Let's try then on comment monitor and now put the mouse over monitor. Yeah.

Timur Khamidulin: Ah, you ah that's what you mean.

Jesus Gonzalez Godoy: So if you see Yeah.

Timur Khamidulin: Okay. Okay. Yeah. Yeah. Yeah. I understand that now. Okay. Okay. The legend. That's what you mean by the

Jesus Gonzalez Godoy: So as the idea is to have this this framework this tool that

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Timur Khamidulin: legend.

Jesus Gonzalez Godoy: others could use then see how you can see here the

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: description.

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: First you have the the signature but also you have the description and the arguments inputs and outputs. It depends of how well you make the dog strings. But now we just uh make a pretty simple uh example, right? But this is the importance. So you can have uh so anyone can have a description. Uh so okay once have that can we go to

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: monitor. Uh also you can click with command and click in

monitor. Yeah. Then if you if you see this is what you put here. Well well cloud create a new directory

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: under with a randomly generated run

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: ID tracks latency persistence time. Okay. can just scroll down and that's it. Okay.

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Jesus Gonzalez Godoy: So basically this is just to um measure the time I think because you just have time it time it this function time it that is pretty commonly in software development but you can in Python it's better to use it as in other forms than in a class. Uh maybe we could see that later. Uh you could use it as for example a decorator. Have you seen decorators? Not right. This is a little bit uh like an advanced concept in Py in

Timur Khamidulin: No.

Jesus Gonzalez Godoy: Python. Uh but when you run some uh methods or your some functions you can uh include this decorator the

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: let's say define this function time it that basically is a timer the time

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: that from start to end from that function but you can use this as a decorator so you don't need to instantiate it and use it uh in every piece of code or remove it of every piece of code.

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Jesus Gonzalez Godoy: So it's easily to manage in another way.

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: Uh but by now let's it's okay then class timer if you see this is kind of things that uh um are well done but are pretty complex too complex. So something interesting about AI is that

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: definitely things different from human. So one uh project or one

code that can be 10 lines maybe could uh create 50 lines instead. And that's because for AI it's pretty easy just to generate code and it's well done. It's not bad code but it's more complex that

Timur Khamidulin: and and and uh I just uh wanted

Jesus Gonzalez Godoy: necessary.

Timur Khamidulin: to ask you like if we speak about the AI uh if uh so if it is uh so Yesterday even we have uh we had a lecture uh the one that was um on um during the daytime uh they were discussing uh so the topic was that uh AI is messing up our code and uh it's messing up I mean by um by making it uh too academic and making it too as you said too uh long uh and uh And you know I

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Jesus Gonzalez Godoy: Mhm.

Timur Khamidulin: was uh wondering whether do you use it in code by

Jesus Gonzalez Godoy: Yeah,

Timur Khamidulin: yourself?

Jesus Gonzalez Godoy: this is a pretty good question. So that's the idea of go deeper in this because uh what I want to show you is that if you just um throw a task to the AI or put a task pretty general task to the AI, it is going to create code but it's not really uh good quality code. uh it's too complex difficult to understand for us maybe easy to understand for AI but difficult to understand for us and some things are without any purpose or so but if you know how to use it and you give just pieces of the task or even if you give all the task but with the uh enough with enough clarity and description and so and well guided the AI is going going to respond well. So answer your question I I encourage the use of

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: AI uh but always uh knowing what the AI is doing right so first you need to be

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Timur Khamidulin: Yes.

Jesus Gonzalez Godoy: pretty uh conscious conscientious

Timur Khamidulin: Yeah. Yeah.

Jesus Gonzalez Godoy: consciousness aware yeah you need to be pretty aware about

Timur Khamidulin: Conscious. Yeah. Yeah. Yeah.

Jesus Gonzalez Godoy: each of the lines that the AI is creating and uh but if you use it

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: correctly it can definitely increase your productivity a lot. See simply like this dock strings I do not put those strings manually uh never I just leave the AI it's pretty good to put those dog strings so I just uh select my code put I want to do strings like uh because you have different uh templates or different standards for doc strings then you can say as uh with inputs, outputs, arguments, examples and so on defined and the output is going to be

Timur Khamidulin: Uh,

Jesus Gonzalez Godoy: awesome.

Timur Khamidulin: and one more question. Do you use any special um, how would I say um, which which AI do you use? Gemini.

Jesus Gonzalez Godoy: Um yes I use Gemini because we have the license here at create but also I like too much

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Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: copilot.

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: Uh yeah in my cell phone I use Gemini for everything. Uh there are many people that is against that.

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: Uh but I think it really increase your productivity too much and also there are a lot of developers that

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: always are saying that um AI developing is stinking also it's pretty bad and but it is just a tool. It's uh it makes what you want right? So, so everything for me everything depends of the input.

Timur Khamidulin: Exactly.

Jesus Gonzalez Godoy: Same rule as the machine learning garbage input garbage output. So it is not so marvelous if you just ask things that

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: you are not not really understand but you if you are developing

also for prompting if you went through pro prompting there is one thing called change of thoughts. Um did you study that change of

Timur Khamidulin: change of thoughts.

Jesus Gonzalez Godoy: thoughts?

00:29:04

Timur Khamidulin: It's uh

Jesus Gonzalez Godoy: So change of thoughts is for a prompt. Maybe if you put uh a task to compiler and say uh can you generate a formula for uh I'm going to say something that has no sense but just to simplify

Timur Khamidulin: Yeah. Yeah.

Jesus Gonzalez Godoy: uh a formula for the relativity theory let's say

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: so maybe Copilot is going to fail uh making mats or something. But if you say hey copilot this is my problem and this is the formulas I have actually can you explain me this formula. Okay now this formula. Okay. Now this okay now how can I combine it? Now how can this explain relativity? Then it is better structure and this is change of thoughts first one basic then uh another prompt based on that prompt and so on until get the final result then um AI is going to answer pretty well. So I use that a lot.

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Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: Yeah. And and finally I think also we are different as AI is different and different AI uh I tested and you can't use the same prompt even for recommendations or even for how they are trained right so maybe you

Timur Khamidulin: Yeah. Yeah.

Jesus Gonzalez Godoy: prefer change of thoughts and Maybe um OpenAI uh makes CHP respond better to change of thoughts but maybe you prefers generalize uh so just give pre- general tasks and maybe uh cloud or uh gemini is done for that general general ideas so different LLMs different prompts

Timur Khamidulin: Yeah, that's that's definite and even what I understood different uh different tasks uh different uh also LLMs uh

Jesus Gonzalez Godoy: Mhm. Also, yeah. So, there is a whole world that many people

do not understand and people

Timur Khamidulin: Yes.

Jesus Gonzalez Godoy: that understand that it's going to uh increase productivity and stay ahead of the others. I think yeah that that's why I

Timur Khamidulin: Okay. Okay.

Jesus Gonzalez Godoy: don't told you write everything by yourself uh but it's important that you understand each piece code and in this case

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Timur Khamidulin: Yeah. Yeah. Yeah. Mhm. Mhm.

Jesus Gonzalez Godoy: this this is too much too much complex but yeah okay so let's let's

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: keep advancing um so where is the error here I think uh it the error me can can we go to the error the compressing data. Uh this is in some place, right? And you just copy paste this.

Timur Khamidulin: You come please.

Jesus Gonzalez Godoy: Or I think is this takes because monitor is pretty small. Basically you just assign this run id the path uh make deer uh create this dictionary then in time it you even don't even use this time it function right it is defined but it's not just uh cannot find root key and you scroll down in monitor class. Monitor class. Yeah. Yeah. This is just the timer, but you didn't use the timer. Uh uh can you

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: go again to the code please to the main function?

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Jesus Gonzalez Godoy: The main function. I think I think this should not fail. Can you save and run again? Okay, this don't fail. So, it's not monitor confused. Then in vector database,

Timur Khamidulin: the

Jesus Gonzalez Godoy: I think that is the error. There is error. Can you comment vector data?

Timur Khamidulin: whole vector database.

Jesus Gonzalez Godoy: No, no, just the next line where you create the vector just to the book.

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: What is happening? Uh, you need to save the

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: script. Okay, next slide please. So with this I think you create a vector database and give the monitor as input. Then we're going to update with the data and there is the error. So let's try save and run again please. Yes, there is there. Now, can you uh put your mouse over vector database? Yeah. So, you have not any dog string.

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Jesus Gonzalez Godoy: You just can see the sign network. See the so for maybe for Monday you can

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: include dogs strings for everything on Monday. for today. Let's see where there is uh overlap because it if the vector database creation is failing you cannot continue with the rack, right?

Timur Khamidulin: uh as maybe I as I understood maybe it's the issue with one

Jesus Gonzalez Godoy: I mean,

Timur Khamidulin: of the PDF files that's why it's um it's um um it's it's with the issue with the PDF. So some PDFs are going through and some are not going through.

Jesus Gonzalez Godoy: I think that the error is because you already has this document in the database and you are indexing it again something like that. Um so can you um this copy paste this while the compressing data or this is a just the compressing data.

Timur Khamidulin: and so what?

Jesus Gonzalez Godoy: Oh. Oh. Can you scroll down in the code?

00:37:47

Timur Khamidulin: No,

Jesus Gonzalez Godoy: In the code. Yeah. Uh, slowly det. Scroll down. Scroll down. Scroll down. here. Update. Scroll down. Scroll down. Okay. So um can we go to main I don't remember how the method was called update database. Okay let's go to that method control and click update database. Mhm. So it is great. So here with self this

with self monitor time it what is what is what is doing here is that create this timer with the name in just milliseconds and just measure the time for this this piece of code load pages Um so first load pages. What is load pages? Can you go to load pages? Okay. Now let's go back to to date and just put there a print like pages loaded successfully or something like that.

00:40:04

Timur Khamidulin: Uh

Jesus Gonzalez Godoy: Uh before chunks

Timur Khamidulin: where

Jesus Gonzalez Godoy: this

Timur Khamidulin: so update successfully.

Jesus Gonzalez Godoy: yes something about but in a print. Yeah, that is okay. Also for prints and things like that, AI is pretty straightforward for completion. I use a lot for completion.

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: Not not for create code, but for completion. Yes.

Timur Khamidulin: Mhm. And so here we write update.

Jesus Gonzalez Godoy: Yeah, that is okay. That's okay.

Timur Khamidulin: It's okay.

Jesus Gonzalez Godoy: Yeah,

Timur Khamidulin: Okay. So that's okay.

Jesus Gonzalez Godoy: it's Yeah,

Timur Khamidulin: Wait, you mean the completion? Okay. Or what?

Jesus Gonzalez Godoy: the completion.

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: Okay, now let's run again.

Timur Khamidulin: Not

Jesus Gonzalez Godoy: Now we have uh different ways to the VA program.

Timur Khamidulin: yet.

Jesus Gonzalez Godoy: One is this. This is the more simple way to see where the problem is. I just go through each line and see where it is reaching the error.

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Jesus Gonzalez Godoy: Um you need to save and run again. I need to say yes. And here you are within the loop. So you need to go out control X or something like that. Oh, you're in Python. Is exit. Yeah. Is you you exit Python with exit.

Timur Khamidulin: Okay. And I now I need to

Jesus Gonzalez Godoy: So I was saying that there is three ways of uh the book. There are there are many ways but one this is the simplest just print uh but it's not so efficient or informative. The other one could be use the visual studio the booger but you need to know how to use it uh if you want to go to that later. And the last and the other one could be used at the booger framework of Python that is also pretty useful. It says uh how many variables you have, how many instances, which value is in each instance and each variable.

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Jesus Gonzalez Godoy: So if you want to debug that those are better options.

Timur Khamidulin: Oh no, I exited the wait. Oh no, wait. Uh virtual environment. I I exited the virtual environment and in 58 I remember. Yes. And uh I command

Jesus Gonzalez Godoy: You just need to see if you are in the correct folder. I think yes. Just Python and T

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: frame.py.

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: So we can see.

Timur Khamidulin: uh interactive. I need the three interactive one. So

Jesus Gonzalez Godoy: Uh, you have the parameters.

Timur Khamidulin: yeah,

Jesus Gonzalez Godoy: I forget that.

Timur Khamidulin: goodbye.

Jesus Gonzalez Godoy: Okay, parsing objects. So at the end you have this loaded 101 pages from data. Can you see that?

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: But the error is before that. So the error is inside this function load pages. So let's go inside inside that function. Can you see why the error is inside that function?

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Timur Khamidulin: Uh so

Jesus Gonzalez Godoy: I mean because all

Timur Khamidulin: error

Jesus Gonzalez Godoy: the

Timur Khamidulin: error. Oh yeah.

Jesus Gonzalez Godoy: all the errors are coming before the the message.

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: So the first function that we track is this update database. We put this because we thought there was in this uh

Timur Khamidulin: Yeah, yeah,

Jesus Gonzalez Godoy: function.

Timur Khamidulin: yeah,

Jesus Gonzalez Godoy: This is correct. So now go inside this function not pages

Timur Khamidulin: yeah.

Jesus Gonzalez Godoy: control and click.

Timur Khamidulin: Long P

Jesus Gonzalez Godoy: So as you can see with self monitor this is just to yeah measure the time that

Timur Khamidulin: Yeah. Yeah. Don't understand. Yeah. Where where we made the mistake.

Jesus Gonzalez Godoy: okay load pages

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: now you have the path in fruit not exist but not found we do not have that error so it is existing define page just errors. So then you make a four in root ether.

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Jesus Gonzalez Godoy: So you go through all the folders in the root and if if not topic there if not if not if is not a directory continue topic equal to topic directory name.

Timur Khamidulin: Interesting.

Jesus Gonzalez Godoy: This is the directory. So if you f if you have a file in the root in

data folder it just continue. But if it is a directory then you go and for each uh file you do something. So here is my first question. So you created this also with AI right?

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: So for example see here it says for PDF path that is the file in the folder right topic

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: is the the folder like GCP git or ra this is going to be topic and then you have

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: for PDF in this part in sorted why this is sorted it. I mean, I do not have a reason why you want to this to be sorted. Do you know why?

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Timur Khamidulin: Um well uh cuz uh because of the order cuz um um without um sorter um

Jesus Gonzalez Godoy: I mean what this globe function

Timur Khamidulin: um

Jesus Gonzalez Godoy: do

Timur Khamidulin: What? Which function?

Jesus Gonzalez Godoy: globe topic dear globe what this does is returns a list of all the files with this pattern that is all the PDFs found in this in this topic directory. Right?

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: So if we have a list and we want to put all the PDFs in the vector database or to whatever, I don't know why they need to be sorted, but let's continue. Maybe for some reason, but this could be also in the first four we have sorted. What's what sorted do is sort alphabetically, right? So there there is no reason to to order or sort this

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: list, neither the um folders list or directories list, neither the files list. Then you try and it says for page for every page in this by PDF loader I think loads the page then you're going to have the text the source the topic and the page an element as dictionary Sorry.

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Jesus Gonzalez Godoy: If there is an an exception, you just put there is an exception. Can you scroll down a little bit? Then fort inserted. Uh I just scroll up scroll up uh because we have txt part. Can you scroll up a little? This uh can you see this variable txt path? Right.

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: So our intention is not indexed any txt or MD files, right? Just PDFs.

Timur Khamidulin: just yes.

Jesus Gonzalez Godoy: So also this is uh extra code.

Timur Khamidulin: So there's no need in this. Yeah, they

Jesus Gonzalez Godoy: So the first the first block is to put PDFs.

Timur Khamidulin: t

Jesus Gonzalez Godoy: The second one is to look for txt and MD files. Uh can you scroll down and if errors? So please just put um a print here before return. Yeah, something like uh before return on date process or something like that

Timur Khamidulin: Print

Jesus Gonzalez Godoy: before return or something like that just to

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Timur Khamidulin: return

Jesus Gonzalez Godoy: identify

Timur Khamidulin: or or or something easier.

Jesus Gonzalez Godoy: Yeah. No, just put something like uh before return. Yeah, before return the return instruction. Yeah, that's it. Now you can save and run again. Mhm. Now you can see here before return. So we know everything is inside here.

Timur Khamidulin: Yep.

Jesus Gonzalez Godoy: Right. So this txt and MD we can remove all that

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: block. So we can go no scroll up.

Timur Khamidulin: Yes. Here and then the

Jesus Gonzalez Godoy: Yeah.

Timur Khamidulin: block.

Jesus Gonzalez Godoy: So all that code is useful as I mentioned before but not useful for us. Right?

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: And then here the error is basically here inside this four cycle. So you can just print put a print before this four. No, leave this. Leave that one. Just create a new print.

00:55:35

Jesus Gonzalez Godoy: Just type print Yeah, but we know their content. Uh yeah, that is correct. No, but uh let's

Timur Khamidulin: It's fine. Okay.

Jesus Gonzalez Godoy: Yeah. Now run. Mhm. So now we can see here before topic kit before topic rack and before topic GCP. So we know better we have a better idea where the error is right.

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: Where is the error?

Timur Khamidulin: So the error is uh in uh uh inside here in the PDF

Jesus Gonzalez Godoy: Yeah, but what kind of PDFs are creating the

Timur Khamidulin: load.

Jesus Gonzalez Godoy: error the GCP folder, right?

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: Because you can see here that all the errors

Timur Khamidulin: Cuz Yeah.

Jesus Gonzalez Godoy: are Yeah,

Timur Khamidulin: Yeah. underneath the GCP.

Jesus Gonzalez Godoy: exactly. Then lastly, we can see here that um you just load uh every page of each PDF file. So what we can do is uh what if we can you remove the three PDF and then put one by one.

00:57:47

Timur Khamidulin: Uh, yep.

Jesus Gonzalez Godoy: So just uh yeah use another temporary folder. also the other one. First, let's leave the folder the folder empty.

Timur Khamidulin: Okay.

Jesus Gonzalez Godoy: Yeah. Now run again. Okay, remove the error. Um, not all the

errors. This ignoring pointing object level in Git. There is another error. So, first uh let's just put one of the PDFs. Okay. So, just remove that and put another one,

Timur Khamidulin: That's

Jesus Gonzalez Godoy: please. Mhm.

Timur Khamidulin: okay. This one is good.

Jesus Gonzalez Godoy: So, the other is the problem. Now if we go to the code no we are

Timur Khamidulin: What about the third one? Don't need to

Jesus Gonzalez Godoy: pretty sure that is the a complete

Timur Khamidulin: check.

Jesus Gonzalez Godoy: guide. Yeah that is the problem. So now inside the the code we can see that we put page page content PDF path name topic and metadata.

01:00:05

Jesus Gonzalez Godoy: So return the problematic PDF please to the to the folder.

Uhhuh. Now we can see that in the error it says found zero objects within object 7075 whereas I don't know what expected compressing data so I am pretty sure that the error is the ladder by PDF flu uh comment that for cycle.

Timur Khamidulin: comment. What?

Jesus Gonzalez Godoy: Uh let me see because you this this for loop but you have that for loop inside a try and exception but you do not print the exception. I mean can you scroll down a little bit? Errors append if errors write JSON just errors. This is another of the complicated things because you see saves errors type. Okay, just instead or or before errors happen, just print the same line. Yeah, that's it. Correct. Now run again. Mhm. Now we can see error loading that PDF because cannot find group object in PDF. So for some reason this PDF is built in a different way of the others.

01:02:29

Jesus Gonzalez Godoy: Uh I do not have rune root sorry.

Timur Khamidulin: Mhm.

Jesus Gonzalez Godoy: Um so maybe you can just ask cloud about this error just to finish. Um, let me Okay. So for example, uh this is one this is a good example. You just copy paste the error, right? But you and cloud here I'm not sure but

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: maybe it is aware of your code and what you are using and so on right so it's correct but when you use uh another like gemini for example that has no context maybe you first put context and

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: why you and what answer do you want? So maybe explain this PDF loader what does what what is the root what why is uh causing the error and how can I fix

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: it right but in this case okay

Timur Khamidulin: Yeah. I I just didn't do it.

Jesus Gonzalez Godoy: um okay okay so what it

01:04:10

Timur Khamidulin: Yeah. Because it's in it's in context. Yeah. But I can do it

Jesus Gonzalez Godoy: says basically I think that

Timur Khamidulin: right.

Jesus Gonzalez Godoy: the PDF is uh incorrectly the build. So it's corrupt, right?

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: Um so in that case I think you can go repair the PDF or maybe use another kind of uh correction for the PDF or things like that. But as this is out of the scope of our project, you just remove that and look another one.

Timur Khamidulin: Yeah. Yeah, I got it.

Jesus Gonzalez Godoy: But we understand.

Timur Khamidulin: Yeah, the corrupted file.

Jesus Gonzalez Godoy: Yeah.

Timur Khamidulin: So, yeah.

Jesus Gonzalez Godoy: But we was able to understood the error. Right.

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: So uh what do you think Tim if for Monday or I mean we can make a little shorten the next project the but it's important that maybe you go through the code

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: and understand each of the lines use cloud.

01:05:27

Timur Khamidulin: Yeah. Yeah. I actually do I actually do this all the time and uh yeah cuz uh

Jesus Gonzalez Godoy: Yeah. Okay.

Timur Khamidulin: here Yeah. Here I see I miss miss some parts but yeah I'll do it by by Monday.

Jesus Gonzalez Godoy: Yeah,

Timur Khamidulin: Uh I'll do Yes.

Jesus Gonzalez Godoy: clean the code. Clean the code if possible.

Timur Khamidulin: Yes.

Jesus Gonzalez Godoy: Make it work as a framework,

Timur Khamidulin: Yes.

Jesus Gonzalez Godoy: not as an interactive application. And we can uh check again.

Timur Khamidulin: Yes.

Jesus Gonzalez Godoy: Uh because next is play a little bit why it is doing short answers. Uh try to tricky with the prompt. uh try to make function the the rules you add in the prompt and check the W and you didn't have the evaluation you told me right you're working on that okay and also check

Timur Khamidulin: Yeah, I'm almost finished. Yeah.

Jesus Gonzalez Godoy: that so I think we can uh leave it the here today and see you on

01:06:35

Timur Khamidulin: Yeah. Yeah. On in Monday after after the after the lecture,

Jesus Gonzalez Godoy: Monday.

Timur Khamidulin: not the lecture, you know, after the one that you told me to go on uh about the drug.

Jesus Gonzalez Godoy: It's It's on Monday.

Timur Khamidulin: Yeah.

Jesus Gonzalez Godoy: Okay. Okay. Yeah. Good.

Timur Khamidulin: Let me let me let me double check.

Jesus Gonzalez Godoy: After what time is it?

Timur Khamidulin: Yes. Internal grid dynamics 11 a.m.

Jesus Gonzalez Godoy: Uh, okay. 11.

Timur Khamidulin: So it's not very convenient for you. Yeah.

Jesus Gonzalez Godoy: Let me let me see. Maybe let let's for now let's put that 11:00

a.m.

Timur Khamidulin: I I can do 11:00 a.m.

Jesus Gonzalez Godoy: It's okay.

Timur Khamidulin: Uh or we can do evening. If you can do evening, I can do late evening, for example, because I can I work sometimes in the office until 8:00 p.m. my time. So, it's 7:00 p.m. your time. So, if you're okay with working in the evening, we can have a call in the evening.

Jesus Gonzalez Godoy: Okay. Okay. Good. So, let me see on Monday and if and I send the invite for 11 or evening. Okay. Thank you.

Transcription ended after 01:08:28

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